Fiscal Year:	FY 2004	**Task Last Updated:**	FY 05/13/2004
PI Name:	Wayner Jr., Peter Ph.D.		
Project Title:	R & D work on the Constrained Vapor Bubble System for a Microgravity Experiment		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Interfacial phenomena		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Troy	**State:**	NY
Zip Code:	12180	**Congressional District:**	21
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	01/01/2001	**End Date:**	04/21/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	ISS		
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:	Wayner, Jr. Peter C.		
COI Name (Institution):	Plawsky, Joel (Rensselaer Polytechnic Institute)		
Grant/Contract No.:	NAG3-2383		
Performance Goal No.:			
Performance Goal Text:			
	A near-term flight experiment concept was developed to pursue both equilibrium and non-equilibrium testing of the CVBHX operation. Flight-qualified apparatus and optical instrumentations are being designed and developed for a flight experiment on the International Space Station in 2005. The microscopic intermolecular force (pressure) field, which is a function of the liquid film thickness profile, was measured using microcomputer enhanced video microscopy based on interferometry. The augmented Young-Laplace equation over a large range of film thickness was evaluated. The overall heat conductance of the CVBHX as a function of liquid volume and heat flow rate was also determined. Models of the transport processes in the contact line region of a CVBHX which include the effects of liquid-solid and liquid-vapor intermolecular forces have already been developed. As part of this effort, these models were further refined and the transport characteristics of the CVBHX were obtained by comparing the experimental data to numerical solutions of the model. The vertical CVBHX was found to be an ideal experimental setup in which to study the dropwise condensation		

Task Description:	process and spreading using ethanol. Therefore, basic research on the region of dropwise condensation is also being emphasized. The objective of this effort is to better understand the fundamental physics of evaporation and condensation as they affect the heat transfer processes in a constrained vapor bubble heat exchanger (CVBHX). This CVBHX consists of a small enclosed container partially filled with a liquid. When a temperature difference is applied to the ends of the CVBHX, evaporation occurs at the hot end and condensation at the cold end resulting in a very effective heat transfer device with great potential in space applications. The use of interfacial phenomena leads to a more natural energy transfer process. This study is synergistic in that the space program needs passive systems like the CVBHX for thermal control and research on a low capillary pressure CVBHX requires a microgravity environment. This study is also multifaceted: (1) It is a basic study in interfacial physics because the CVBHX is used to measure the interfacial intermolecular force field; (2) It is a basic study in fluid physics because it evaluates fluid flow in ultra-thin liquid films; (3) It is a dynamic study in thermal transport because it evaluates the heat transfer capabilities of a small scale heat exchanger; and (4) The CVBHX can also be called a wickless heat pipe heat exchanger. The results from this study are not only applicable to space applications, but can also help the design of conventional heat pipes which are poorly understood.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Work continues on the development of the flight experiment hardware, the flight experiment software, and on ground-based experiments to support the flight. Efforts have lately focused on the stability characteristics of the CVB device. Since the beginning of this contract we have made 16 presentations and we have published 16 proceedings, peer-reviewed papers and abstracts. Copies of these publications can be obtained from the PIs.
Bibliography Type:	Description: (Last Updated: 09/25/2012)
Articles in Peer-reviewed Journals	Gokhale SJ, Plawsky JL, Wayner PC, DasGupta S. "Inferred pressure gradient and fluid flow in a condensing sessile droplet based on the measured thickness profile." Physics of Fluids. In press, April 2004. , Apr-2004
Presentation	Gokhale, S.J., DasGupta, S., Plawsky, J.L., and Wayner, P.C., Jr. "Experimental Evaluation of Interfacial Prile and Pressure Variation in a Spreading Drop During Condensation" AIChE Annual Meeting, San Francisco Abstract #244d Nov-2003